Metriplectic Completion of the Frohmanian Symplectic Tether: Algebraic Structure, a Finite-Dimensional Viscous Model, and a Lean 4 Formalization

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Abstract

We study how the Frohmanian tether kernel, a symmetric quadratic correction to the Arnold Lie–Poisson bracket on divergence-free vorticity fields on $\mathbb{T}^3$, fits into the metriplectic formalism of Morrison and collaborators. We prove, and check in Lean 4, three facts. (i) For an abstract metriplectic system, energy is conserved and entropy production equals $(S, S) \geq 0$. (ii) Stated against the tether development itself (imported as a pinned Lean dependency, not re-proved), the metric bracket of any metriplectic system on the tether's state space whose energy is the tether's kinetic-energy Hamiltonian vanishes with that Hamiltonian in either slot; if the tether kernel is a summand of such a metric bracket next to a degenerate viscous summand, the tether kernel itself vanishes against the Hamiltonian in both slots; and, because the tether kernel is negative semidefinite on the diagonal, it cannot be the whole metric bracket of a metriplectic system without having zero diagonal and producing no entropy. (iii) We construct a concrete finite-dimensional metriplectic system with viscous dissipation (velocity coefficients $u \in \mathbb{R}^n$ and an entropy coordinate), prove all structure axioms, compute its equations of motion $\dot{u}_i = (J(u)u)_i - \nu_i u_i$, $\dot{s} = \sum_i \nu_i u_i^2$, and give a three-mode triad instance with explicit non-vacuity checks. The finite-dimensional model is not derived from a Fourier truncation of the Navier–Stokes equations, and nothing here recovers or formalizes the continuum Navier–Stokes or Navier–Stokes–Fourier equations. This is not a solution of the Clay Millennium problem; a kernel check is not a geometric or physical proof.

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1 Introduction

The Euler equations of ideal incompressible hydrodynamics are Lie–Poisson dynamics on the dual of the Lie algebra of divergence-free vector fields (Arnold, 1966). Adding viscosity gives the Navier–Stokes equations, which are dissipative. The metriplectic formalism [2, 4] adds a symmetric positive-semidefinite *metric bracket* to a Poisson bracket so that energy is conserved and entropy does not decrease.

The Frohmanian tether kernel T [7] is a symmetric correction to the Arnold bracket. Its Lean development records an obstruction: if T were degenerate on the kinetic-energy Hamiltonian (condition (C2)), symmetry would force $T(H, M) = 0$ for every M, so the tether would supply no damping along the energy direction. Condition (C2) is *not* proved for T in that development.

This paper asks what the metriplectic formalism says about the tether, and records exactly what has been checked by the Lean kernel. Section 8 lists the Lean declarations and how they were audited. The scope is algebraic and finite-dimensional; continuum statements in Sections 2–3 are background, cited from the literature, and are not formalized.

2 Arnold Lie–Poisson structure (background)

Let $\mathfrak{g} = \mathfrak{X}_{\mathrm{div}}(\mathbb{T}^3)$ be the Lie algebra of smooth divergence-free vector fields on the three-torus, with dual identified with vorticity fields. The Lie–Poisson bracket of two functionals of the vorticity ω is

$$\{F, G\}(\omega) = \int_{\mathbb{T}^3} \omega \cdot \left(\frac{\delta F}{\delta \omega} \times \frac{\delta G}{\delta \omega} \right) dx, \tag{1}$$

and the kinetic energy $H(\omega) = \frac{1}{2} \int_{\mathbb{T}^3} |u|^2 \, dx$, $u = \mathrm{curl}^{-1}\omega$, generates the Euler equations $\partial_t \omega + \mathcal{L}_u \omega = 0$. This section is classical background and is not formalized here.

3 Metriplectic systems

Definition 3.1 (Metriplectic system)**.** *A metriplectic system on a state set X consists of a Poisson bracket $\{\cdot,\cdot\}$ and a metric bracket $(\cdot,\cdot)$ on functionals $X \to \mathbb{R}$, an energy H and an entropy S, such that for all functionals F, G and all states:*

$$\{F, G\} = -\{G, F\} \quad \textit{(skew-symmetry)}, \tag{2}$$
$$(F, G) = (G, F) \quad \textit{(symmetry)}, \tag{3}$$
$$(F, F) \geq 0 \quad \textit{(positive semidefiniteness)}, \tag{4}$$
$$\{F, S\} = 0 \quad \textit{(entropy is a Casimir)}, \tag{5}$$
$$(F, H) = 0 \quad \textit{(energy degeneracy)}. \tag{6}$$

The evolution of an observable is $\dot{F} = \{F, H\} + (F, S)$.

The Jacobi identity for $\{\cdot,\cdot\}$ is not part of this definition and is not used below. The previous version of this paper listed "$(S,H) \geq 0$" as the entropy condition; that was an error, since $(S,H) = 0$ by (6) and (3). The correct condition is (4), which gives $(S,S) \geq 0$.

For the compressible Navier–Stokes–Fourier system, Morrison, Updike, Zaidni and Barham construct the metric bracket from a 4-bracket built from viscosity and conductivity operators [4, 5, 6]. We cite those constructions as background only. **They are not formalized in this work, and no statement here recovers the continuum equations.**

4 Thermodynamic consistency (formalized)

Theorem 4.1 (Energy conservation)**.** *In a metriplectic system, $\dot{H} = \{H,H\} + (H,S) = 0$.*

Proof. By (2) with $F = G = H$, $\{H,H\} = -\{H,H\}$, so $\{H,H\} = 0$. By (3), $(H,S) = (S,H)$, and $(S,H) = 0$ by (6) with $F = S$. □

Theorem 4.2 (Entropy production)**.** *In a metriplectic system, $\dot{S} = \{S,H\} + (S,S) = (S,S) \geq 0$.*

Proof. By (2), $\{S,H\} = -\{H,S\}$. By (5) applied to the functional $F = H$, $\{H,S\} = 0$. Hence $\{S,H\} = 0$ and $\dot{S} = (S,S)$, which is ≥ 0 by (4) with $F = S$. □

Remark 4.3. *The draft Lean proof of Theorem 4.2 cited the Casimir condition at $F = S$, which gives $\{S,S\} = 0$, not the needed $\{S,H\} = 0$, and did not compile. The Lean proof now follows the argument above exactly: Casimir condition at $F = H$, then skew-symmetry.*

Theorem 4.4 (Symmetric swap)**.** *If B is symmetric and $B(F,H) = 0$ for every F, then $B(H,F) = 0$ for every F.*

Proof. $B(H,F) = B(F,H) = 0$. □

5 The Frohmanian tether and the damping obstruction

In the imported Lean development [7], the tether kernel is defined as

$$T(F,G)(\omega) = -\kappa \, \|\omega\|_{L^2}^2 \int_{\mathbb{T}^3} \left(\Pi_u \tfrac{\delta F}{\delta \omega}\right) \cdot \left(\Pi_u \tfrac{\delta G}{\delta \omega}\right) dx, \qquad \|\omega\|_{L^2}^2 = \int |\omega|^2, \tag{7}$$

with the global enstrophy outside the integral, $\kappa > 0$, and Π_u the L^2-orthogonal projector onto the complement of $\mathrm{span}\{u\}$, u the velocity recovered from ω. The previous version of this paper wrote a pointwise weight $|\omega(x)|^2$ inside the integral; that is a different object and is not the one formalized. The three conditions are:

(C1) coadjoint invariance — **not proved** for T in that development;

(C2) degeneracy on the kinetic-energy Hamiltonian, $T(F,H) = 0$ for all F — **not proved** for T in that development (an earlier route assumed $\delta H/\delta\omega = u$, which is false with its definitions, and was removed);

(C3) controllable negative feedback, $T(F,F) = -\kappa\|\omega\|^2 \int \|\Pi_u \delta F\|^2$ — proved, as an equality.

Correction to the previous version. The previous version said that "a theorem already verified in Lean" gives $T(M,H) = 0$ for every M. That was a misattribution: $T(M,H) = 0$ for all M *is* condition (C2), a hypothesis. What the Lean development proves is the other slot, $T(H,M) = 0$, *assuming* (C2) (lemma `left_H_zero_of_C2_symm`, specialized in `tetherKernel_left_H_zero`). Since (C2) is not proved for T, neither identity is established unconditionally for T.

6 The obstruction stated against the imported kernel

The Lean file `MetriplecticNS/TetherBridge.lean` imports the tether development and works on its state space (`CoadjointOrbit`) with its Hamiltonian (`KineticEnergyHamiltonian`).

Theorem 6.1 (Metric obstruction)**.** *Let $(\{\cdot,\cdot\},(\cdot,\cdot),H,S)$ be a metriplectic system on divergence-free vorticity fields with H equal to the tether's kinetic-energy Hamiltonian. Then the metric bracket satisfies (C2), and $(H,M)=0$ for every functional M.*

Proof. (C2) for the metric bracket is (6). The imported lemma `left_H_zero_of_C2_symm`, applied to the metric bracket with symmetry (3) and this (C2), gives $(H,M)=0$. □

Theorem 6.2 (Tether summand)**.** *In the setting of Theorem 6.1, suppose $(F,G)=V(F,G)+T(F,G)$ for some bracket V with $V(F,H)=0$ for all F. Then T satisfies (C2), and $T(H,M)=T(M,H)=0$ for every M.*

Proof. For every F: $0=(F,H)=V(F,H)+T(F,H)=T(F,H)$, which is (C2) for T. The imported theorem `tetherKernel_left_H_zero` then gives both identities. □

So, under the hypotheses of Theorem 6.2, the tether summand does not couple to the energy direction at all. Theorem 6.2 does not prove (C2) for T; it shows that (C2) for T is *forced* by placing T in such a split metric bracket.

Lemma 6.3 (Sign of the tether diagonal)**.** *$T(F,F)(\omega)\le 0$ for every F and ω.*

Proof. By (C3), $T(F,F)(\omega)=(-\kappa)\cdot\|\omega\|^2\cdot\int\|\Pi_u\delta F\|^2$. Here $-\kappa\le 0$ since $\kappa>0$; $\|\omega\|^2=\int|\omega|^2\ge 0$; and the last integral has a nonnegative integrand, so it is ≥ 0 (in Lean, the Bochner integral of a nonnegative function is nonnegative, including when the function is not integrable and the integral is 0 by convention). A nonpositive number times two nonnegative numbers is nonpositive. □

Theorem 6.4 (Sign obstruction)**.** *If the metric bracket of a metriplectic system on vorticity fields equals T, then $T(F,F)=0$ for all F and all ω, and in particular $\dot S=(S,S)=0$: no entropy is produced.*

Proof. (4) gives $T(F,F)\ge 0$; Lemma 6.3 gives $T(F,F)\le 0$. Then $\dot S=(S,S)$ by Theorem 4.2, and $(S,S)=T(S,S)=0$. □

Thus a positive entropy production cannot come from the tether kernel. In a split $(\cdot,\cdot)=V+T$, $(S,S)=V(S,S)+T(S,S)\le V(S,S)$: the tether can only lower the entropy production supplied by V, and V must dominate it for (4) to hold. These statements replace the previous version's claim that "the tether kernel may be inserted as one summand of the metric bracket" and that "energy conservation continues to hold because T is already degenerate on H"; that degeneracy is not proved.

7 A finite-dimensional viscous metriplectic system

This section is the concrete instance formalized in `MetriplecticNS/Galerkin.lean`. It is a genuine metriplectic system, but it is finite-dimensional, and it is not derived from the Navier–Stokes equations.

7.1 Definition

Fix $n \in \mathbb{N}$. The state is $x = (u, s) \in \mathbb{R}^n \times \mathbb{R}$ (u are velocity coefficients, s is an entropy / internal-energy coordinate). For a functional $F : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}$ let $\partial_{u_i} F(x) = DF(x)[(e_i, 0)]$ and $\partial_s F(x) = DF(x)[(0, 1)]$, where $DF(x)$ is the Fréchet derivative (taken to be 0 where F is not differentiable, Mathlib's convention, so that the brackets are defined on every functional). The data are a map $J : \mathbb{R}^n \to \mathbb{R}^{n \times n}$ with $J(u)^{\mathsf{T}} = -J(u)$ for every u, and viscosities $\nu_i \geq 0$. Define

$$\{F, G\}(x) = \sum_{i,j} \partial_{u_i} F \; J(u)_{ij} \; \partial_{u_j} G, \qquad (PF)_i(x) = \partial_{u_i} F - (\partial_s F)\, u_i,$$

$$(F, G)(x) = \sum_i \nu_i \, (PF)_i \, (PG)_i, \qquad H(u, s) = \tfrac{1}{2}\|u\|^2 + s, \qquad S(u, s) = s.$$

(For a Fourier truncation one would take $\nu_i = \nu |k_i|^2$; we do not need that interpretation.)

7.2 Derivatives

Lemma 7.1. *For every state $x = (u, s)$ and index i: $\partial_{u_i} H = u_i$, $\partial_s H = 1$; $\partial_{u_i} S = 0$, $\partial_s S = 1$; for $K = \tfrac{1}{2}\|u\|^2$, $\partial_{u_i} K = u_i$, $\partial_s K = 0$; for the coordinate $c_j(x) = u_j$, $\partial_{u_i} c_j = \delta_{ij}$, $\partial_s c_j = 0$. Consequently $(PH)_i = 0$, $(PS)_i = -u_i$, $(PK)_i = u_i$, $(Pc_j)_i = \delta_{ij}$.*

Proof. $x \mapsto \|u\|^2$ has derivative $v \mapsto 2\langle u, v_1 \rangle$ (chain rule: $\|\cdot\|^2$ has derivative $2\langle u, \cdot \rangle$ and $x \mapsto u$ is the continuous linear projection). Hence $DK(x)[v] = \langle u, v_1 \rangle$ and $DH(x)[v] = \langle u, v_1 \rangle + v_2$, since $x \mapsto s$ is the continuous linear projection onto the second factor. Evaluating at $v = (e_i, 0)$ gives $\langle u, e_i \rangle = u_i$; at $v = (0, 1)$ gives 0 for K and 1 for H. S and c_j are continuous linear maps, so they are their own derivatives: $DS(x)[v] = v_2$ and $Dc_j(x)[v] = (v_1)_j$. The values of P follow from the definition, e.g. $(PH)_i = u_i - 1 \cdot u_i = 0$. □

7.3 The structure axioms

Theorem 7.2. *The data above satisfy (2)–(6).*

Proof. Skew-symmetry. Exchange the order of summation and use $J_{ji} = -J_{ij}$: $\{G, F\} = \sum_{i,j} \partial_{u_i} G \, J_{ij} \, \partial_{u_j} F = \sum_{i,j} \partial_{u_j} G \, J_{ji} \, \partial_{u_i} F = -\sum_{i,j} \partial_{u_i} F \, J_{ij} \, \partial_{u_j} G = -\{F, G\}$. *Symmetry* of $(\cdot, \cdot)$ is commutativity of the product $(PF)_i (PG)_i$. *Positivity:* $(F, F) = \sum_i \nu_i (PF)_i^2 \geq 0$ since $\nu_i \geq 0$. *Casimir:* $\partial_{u_j} S = 0$ for every j, so every term of $\{F, S\}$ vanishes. *Energy degeneracy:* $(PH)_i = 0$ for every i, so every term of (F, H) vanishes. □

7.4 Equations of motion

Proposition 7.3. *For $\dot{F} = \{F, H\} + (F, S)$:*

$$\dot{u}_i = (J(u)u)_i - \nu_i u_i, \qquad \dot{s} = \sum_i \nu_i u_i^2, \qquad \frac{d}{dt}\tfrac{1}{2}\|u\|^2 = -\sum_i \nu_i u_i^2, \qquad \dot{H} = 0, \qquad \dot{S} \geq 0.$$

Proof. For $F = c_i$: $\{c_i, H\} = \sum_{a,b} \delta_{ia} J_{ab} u_b = (Ju)_i$ and $(c_i, S) = \sum_a \nu_a \delta_{ia} (-u_a) = -\nu_i u_i$. For $F = S$: $\{S, H\} = 0$ since $\partial_u S = 0$, and $(S, S) = \sum_a \nu_a (-u_a)^2$. For $F = K$: $\{K, H\} = \sum_{a,b} u_a J_{ab} u_b = 0$, because the quadratic form $Q = u^{\mathsf{T}} J u$ of a skew matrix satisfies $Q = -Q$ (exchange the summation order and use skew-symmetry); and $(K, S) = \sum_a \nu_a u_a (-u_a)$. The last two claims are Theorems 4.1 and 4.2 applied to the system of Theorem 7.2. □

So kinetic energy lost to viscosity equals entropy gained: $\frac{d}{dt} K = -\dot{S}$.

7.5 A three-mode triad instance and non-vacuity

Take $n = 3$, $\nu_i = 1$, and

$$J(u) = \begin{pmatrix} 0 & u_2 & u_1 \\ -u_2 & 0 & 3u_0 \\ -u_1 & -3u_0 & 0 \end{pmatrix}, \qquad J(u)u = (2u_1u_2,\ 2u_0u_2,\ -4u_0u_1),$$

a quadratic interaction with $2 + 2 - 4 = 0$, of the energy-conserving triad type that appears in Galerkin truncations of fluid equations. Skew-symmetry of $J(u)$ is checked entry by entry. We do not claim the Jacobi identity for this J, and we do not claim this triad is a Fourier truncation of Navier–Stokes.

Proposition 7.4 (Non-vacuity)**.** *Let* $x_0 = (e_1 + e_2, 0)$. *Then:* $\{c_0, c_1\}(x_0) = 1 \neq 0$*;* $(c_0, c_0)(x_0) = 1 \neq 0$*;* $H(0,0) = 0 \neq 1 = H(0,1)$*;* $\dot{S}(x_0) = 2 > 0$*; and* $\dot{u}_0(x_0) = 2u_1u_2 - u_0 = 2$*, so the reversible part of the motion is nonzero.*

Proof. In general $\{c_i, c_j\} = J(u)_{ij}$ and $(c_i, c_i) = \nu_i$ by Lemma 7.1. At $u = e_1 + e_2 = (0, 1, 1)$: $J_{01} = u_2 = 1$; $\nu_0 = 1$; $\dot{S} = \sum u_i^2 = 2$; $\dot{u}_0 = 2 \cdot 1 \cdot 1 - 0 = 2$. □

8 Lean 4 formalization and reproducibility

Modules.

- `MetriplecticNS/Basic.lean`: `MetriplecticSystem`; `energy_conserved`, `entropy_production`, `entropy_production_eq`, `tether_damping_obstruction` (Theorems 4.1–4.4).
- `MetriplecticNS/TetherBridge.lean`: imports `FrohmanianTether.Combined`; `metric_degenerate`, `metric_left_H_zero`, `tetherKernel_summand_no_H_coupling`, `tetherKernel_diag_nonpos`, `tetherKernel_as_metric_diag_zero`, `tetherKernel_as_metric_no_entropy_production` (Section 6).
- `MetriplecticNS/Galerkin.lean`: the system `system D`, the five axioms, equations of motion `evolution_coord`, `evolution_S`, `evolution_K`, `kinetic_loss_eq_entropy_gain`, and the triad instance with `triad_poisson_ne_zero`, `triad_metric_ne_zero`, `H_not_const`, `triad_entropy_production_pos`, `triad_evolution_coord0` (Section 7).

Dependency. The tether development is the Lake package `FrohmanianTether` from https://github.com/BenFrohman/frohmanian-tether-sdiff, pinned to commit `8b7bd6741f8a54587db6a95b4a5199dcf2775f97`. Both repositories use Lean `v4.30.0-rc1` and Mathlib commit `0c154d67103f74be3a0f2c509f72ccbf5be9f2a7`, so the kernel lemmas are imported rather than re-proved.

Audit. `scripts/kernel_check.sh` rejects `sorry`, `admit`, `native_decide`, `axiom`, `opaque` and `unsafe` in the sources, and runs `#print axioms` on all 73 declarations of the three modules. Each depends only on `propext`, `Classical.choice` and `Quot.sound` (this output is transitive, so it covers the imported tether lemmas). Continuous integration runs `lake build` and the audit on every push and uploads the audit output.

Rebuild. `lake exe cache get && lake build && ./scripts/kernel_check.sh`. The exact source commit for this version is `b41ff89e58d12a39edcbc5706c795a9aa322dc35`.

What is not formalized. The Arnold bracket on vorticity fields, functional derivatives on Sobolev spaces, any integration by parts, the viscous term $\nu\Delta u$, the 4-bracket constructions, and recovery of the Navier–Stokes or Navier–Stokes–Fourier equations. The previous version said the viscous term was recovered "by integration by parts" in Lean and that the specialization to integral functionals of fluid densities was formalized; neither was true, and neither is claimed now.

9 Conclusion and open problems

The metriplectic formalism sharpens the earlier obstruction. With the tether's own Hamiltonian, any metriplectic metric bracket vanishes against H in both slots; a tether summand next to a degenerate viscous summand is forced to satisfy (C2) and then vanishes against H; and the tether's diagonal has the wrong sign to produce entropy, so positive entropy production must come from another summand, such as a viscous one. A concrete finite-dimensional viscous metriplectic system shows the structure is consistent and non-vacuous.

Open problems:

- prove or refute (C2) for the tether kernel;
- derive a finite-dimensional system of the type of Section 7 from an actual Fourier truncation of the Navier–Stokes nonlinearity, with a proof that the truncated bracket is skew;
- formalize an infinite-dimensional metriplectic structure for the incompressible equations on Sobolev spaces;
- find a non-symmetric or higher-order geometric correction that evades the obstruction while keeping coadjoint equivariance.

Nothing here bears directly on the global regularity of the three-dimensional Navier–Stokes equations.

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